

UNIT 1:

INTRODUCTION TO COMPUTER SYSTEMS

DIGITAL FUNDAMENTALS WITH LOGISIM

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Delivery Rules:

Correction:

-  For each day of delay without justification in delivery, 1 point will be subtracted from the grade.
- The statements of the exercises should appear clearly with the answers to the exercises.
- The answers to questions must be clearly delimited (use **Verdana** font size **12** in blue *italics* – **Answer Style**): *[Answer]*. Answers must be enclosed in square brackets *[]*.
-  It is very important to take care of the presentation and cleanliness of the answers given to the teacher. The letter must be clear and legible.
-  Prepare a report of the actions carried out to solve proposed activities.
-  Tasks must be answered in **English**.
-  In the event that copying or plagiarism is detected with the exercises of other classmates, the exercise mark will be invalidated.

Delivery method:

- Through the Moodle GVA AULES platform: <https://aules.edu.gva.es/>

Documents to deliver:

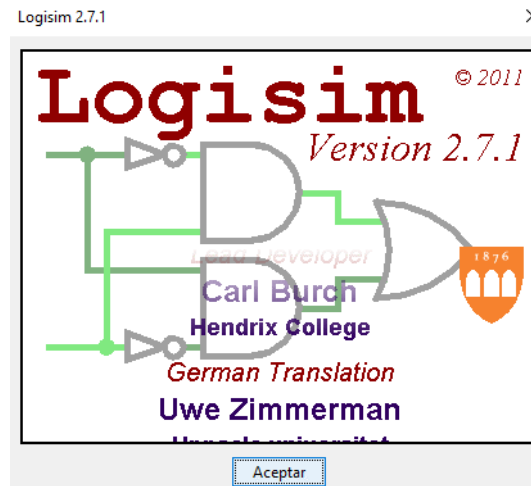
-  It will be possible to use the word processor to prepare the document with the answers.
-  **PDF document format.** Rename the document, and add your last name and first name at the end.
- The suggested name for the **Logism** files is: **ActivityX-<identifier>.circ** (where **identifier** is initials of student's full name followed by the current year). *For example, for activity number 1 of a student named José Gaspar Sánchez García, the file will be called **Activity1-JGSG25.circ**.*
-  Create a compressed package with a **.tgz** extension that contains the files with a **.circ** extension for each of the **Logism projects**.

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1.What is Logisim?

Logisim is an educational tool for designing and simulating digital logic circuits. With its simple toolbar interface and simulation of circuits as you build them, it is simple enough to facilitate learning the most basic concepts related to logic circuits. With the capacity to build larger circuits from smaller subcircuits, and to draw bundles of wires with a single mouse drag, Logisim can be used (and is used) to design and simulate entire CPUs for educational purposes.



[Download Logisim!](#)

The main features of Logisim are:

- It is free! (Logisim is open-source ([GPL](#)).)
- It runs on *any* machine supporting Java 5 or later; special versions are released for MacOS X and Windows. The cross-platform nature is important for students who have a variety of home/dorm computer systems.
- The drawing interface is based on an intuitive toolbar. Color-coded wires aid in simulating and debugging a circuit.
- The wiring tool draws horizontal and vertical wires, automatically connecting to components and to other wires. It's very easy to draw circuits!
- Completed circuits can be saved into a file, exported to a GIF file, or printed on a printer.
- Circuit layouts can be used as "subcircuits" of other circuits, allowing for hierarchical circuit design.
- Included circuit components include inputs and outputs, gates, multiplexers, arithmetic circuits, flip-flops, and RAM memory.
- The included "combinational analysis" module allows for conversion between circuits, truth tables, and Boolean expressions.

2. Remember what you studied.

2.1.1. Boolean Algebra.

Boolean algebra, also called Boolean algebra, in **digital electronics**, computer science, and mathematics is an algebraic structure that schematizes logical operations.

It is named after George Boole (2 November 1815 – 8 December 1864), a self-taught English mathematician, who was the first to define it as part of a logical system, initially in a small pamphlet, **The Mathematical Analysis of Logic**, published in 1847.

Boolean algebra was an attempt to use algebraic techniques to deal with expressions of propositional logic. It was later extended as a more important book known as **An Investigation of the Laws of Thought**, published in 1854.

The respective interpretations of symbols 0 and 1 in the logic system are Nothing and Universe.

Today, Boolean algebra is widely applied in the field of electronic design. Claude Shannon was the first to apply it in the design of bistable electrical switching circuits, in 1948. This logic can be applied to two fields:

- To the analysis, because it is a concrete way of describing how the circuits work.
- To the design, since having a function applies that algebra to be able to develop an implementation of the function.

A Boolean algebra is an algebraic structure called a complementary and **distributive lattice** $(B, +, \cdot)$, for which the following axioms are met.

1. Associative:

$$\forall a, b, c \in B$$

$$(a + b) + c = a + (b + c)$$

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

2. Absorption:

$$\forall a, b \in B$$

$$a \cdot (a + b) = a + a \cdot b = a$$

$$a + (a \cdot b) = (a + a) \cdot (a + b) = a \cdot (a + b) = a$$

3. Commutative:

$$\forall a, b \in B$$

$$a + b = b + a$$

$$a \cdot b = b \cdot a$$

4. Idempotencia:

$$\forall a \in B$$

$$a + a = a$$

$$a \cdot a = a$$

5. Complementation:

$$\exists 0, 1 \in B; \forall a \in B \exists \bar{a} \in B$$

$$a + \bar{a} = 1$$

$$a \cdot \bar{a} = 0$$

6. Distributive:

$$\forall a, b, c \in B$$

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

$$a + (b \cdot c) = (a + b) \cdot (a + c)$$

A Binary or **Bivalent Boolean Algebra** is a Boolean Algebra in which there are only two elements: $B = \{0, 1\}$.

$$\forall a \in B \ a = 0 \vee a = 1$$

$$0+0=0$$

$$0+1=1$$

$$1+0=1$$

$$1+1=1$$

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

In propositional logic and Boolean algebra, **De Morgan's Laws** are a pair of transformation rules that are both valid rules of inference. The norms allow the expression of conjunctions and disjunctions purely in terms of via negation.

$$\forall a, b \in B$$

$$\overline{a \cdot b} = \bar{a} + \bar{b}$$

$$\overline{a + b} = \bar{a} \cdot \bar{b}$$

2.1.2. Logic Gates

A **logic gate** is an electronic device with a Boolean-type function. They make up switching circuits integrated into a **chip**.

The implementation of Boolean functions is based on the **technological capacity** to build electronic devices (logic gates) that execute the operations of Boolean Algebra.

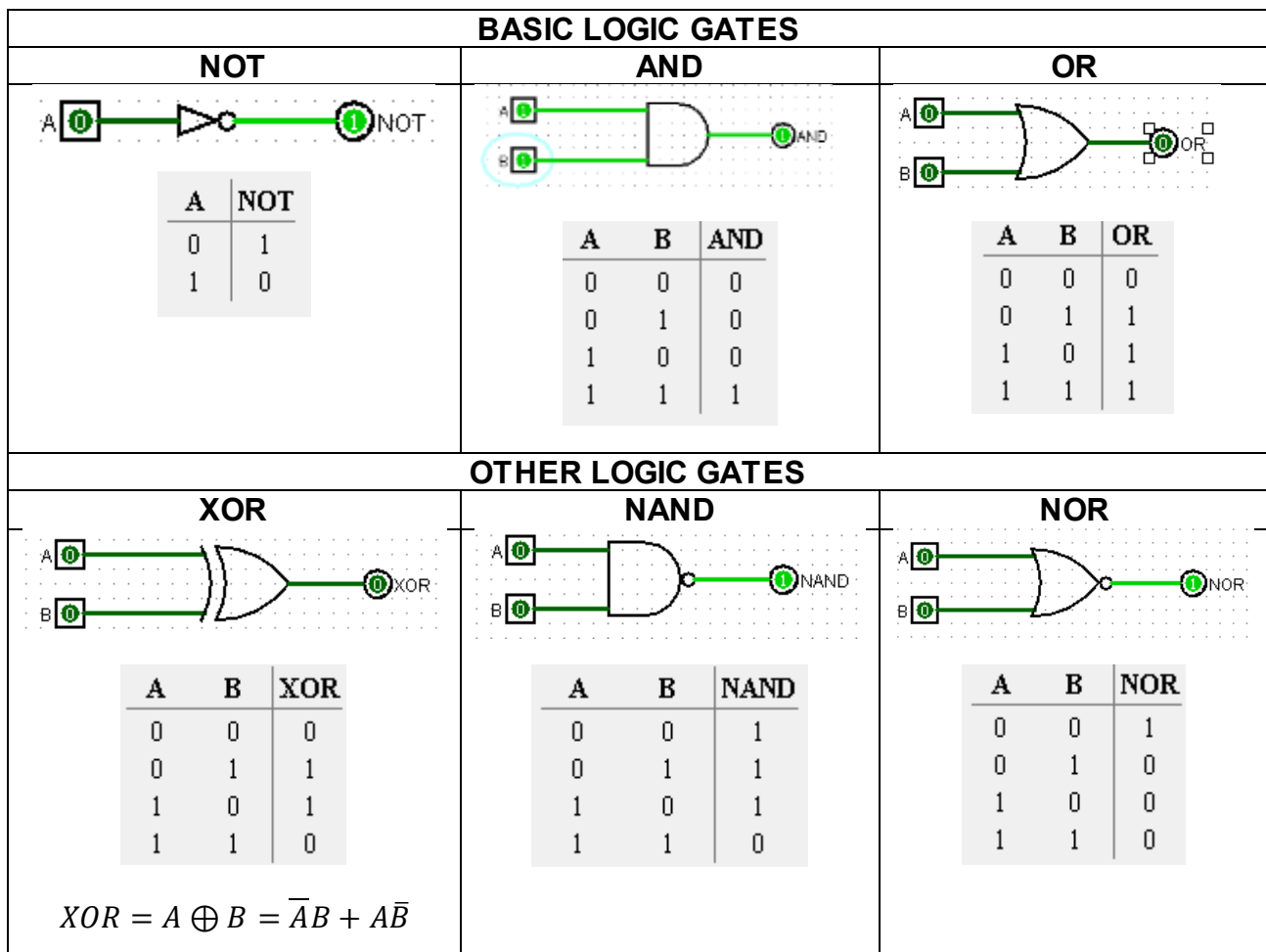


Figure 1. Summary of Logic Gates.

2.1.3. Analysis and synthesis of digital circuits

Analysis: Determine the behavior and functionality of a digital circuit.

Example 1. Analyze the behavior and functionality of the following circuit.

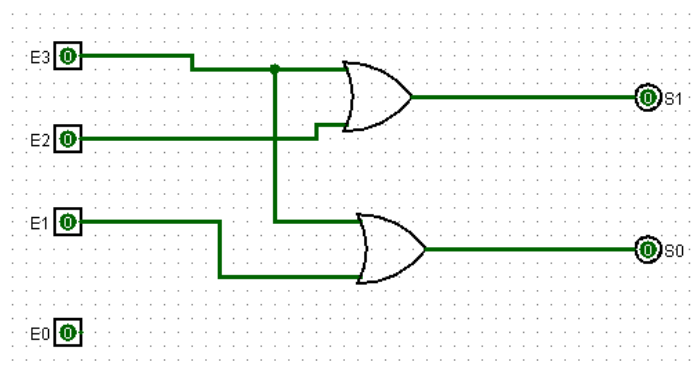


Figure 2. Analysis digital systems.

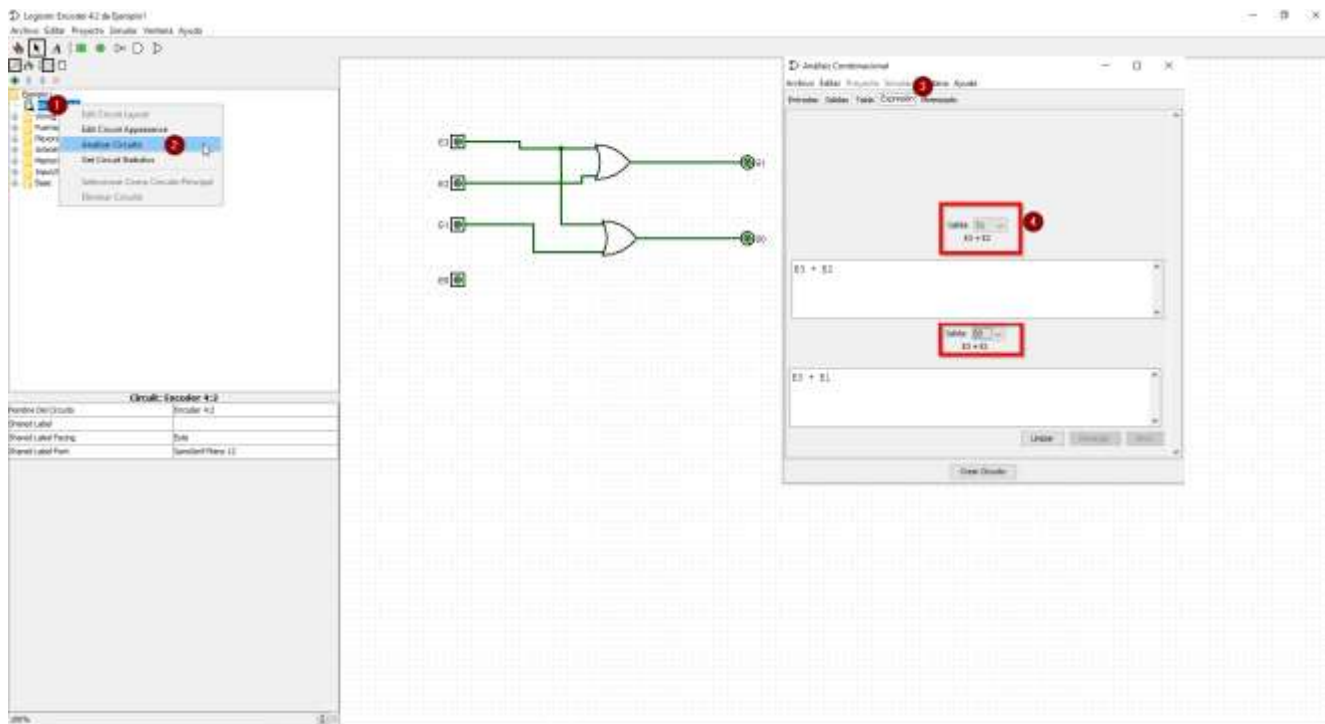


Figure 3. Analyze Circuit >> Combinational analysis.

1. From the circuit we obtain the algebraic expressions which represent the logical functions:

$$S_1 = E_3 + E_2$$

$$S_0 = E_3 + E_1$$

2. Using the output functions, we can obtain the truth table and analyze the behavior of the circuit.

The image shows the 'Análisis Combinacional' window with the 'Table' tab selected. It displays a truth table for the 4:2 encoder circuit. The table has columns for inputs E3, E2, E1, E0 and outputs S1, S0. The first column is a checkbox for selecting rows. The table shows 16 rows of input combinations and their corresponding output values.

	E3	E2	E1	E0	S1	S0
☐	0	0	0	0	0	0
☐	0	0	0	1	0	0
☐	0	0	1	0	0	1
☐	0	0	1	1	0	1
☐	0	1	0	0	1	0
☐	0	1	0	1	1	0
☐	0	1	1	0	1	1
☐	0	1	1	1	1	1
☐	1	0	0	0	1	1
☐	1	0	0	1	1	1
☐	1	0	1	0	1	1
☐	1	0	1	1	1	1
☐	1	1	0	0	1	1
☐	1	1	0	1	1	1
☐	1	1	1	0	1	1
☐	1	1	1	1	1	1

Figure 4. Truth table encoder 4:2.

3. In this particular case the circuit is a 4:2 encoder. Its outputs binary encodes the active input. If there is more than one active input, the output will be indeterminate.

Synthesis: Design the digital circuit from the description of a behavior or functionality.

Example 2. Design and implement a full adder:

The stages to design a logic circuit are:

1. Identify the variables involved in the problem.

Inputs: A, B, Cin.

Outputs: Sum, Cout.

2. Group the identified variables into input variables (independent) and output variables (dependent).
3. Interpret the logical dependency relationships between inputs and outputs and express the functions they represent in a truth table.

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Figure 5. Full adder truth table.

4. Obtain the minimum algebraic expressions of the logical functions that represent the output variables.

$$Sum = A \cdot B \cdot Cin + A \cdot \bar{B} \cdot \bar{Cin} + \bar{A} \cdot B \cdot \bar{Cin} + \bar{A} \cdot \bar{B} \cdot Cin$$

$$Cout = A \cdot Cin + B \cdot Cin + A \cdot B$$

5. Implement with logic gates the functions that solve the problem.

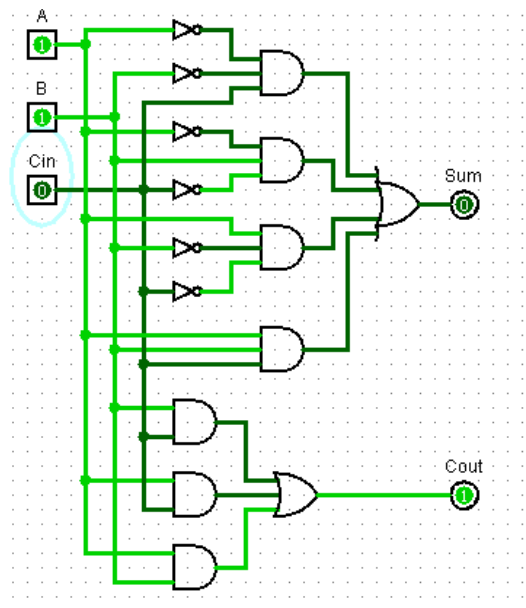
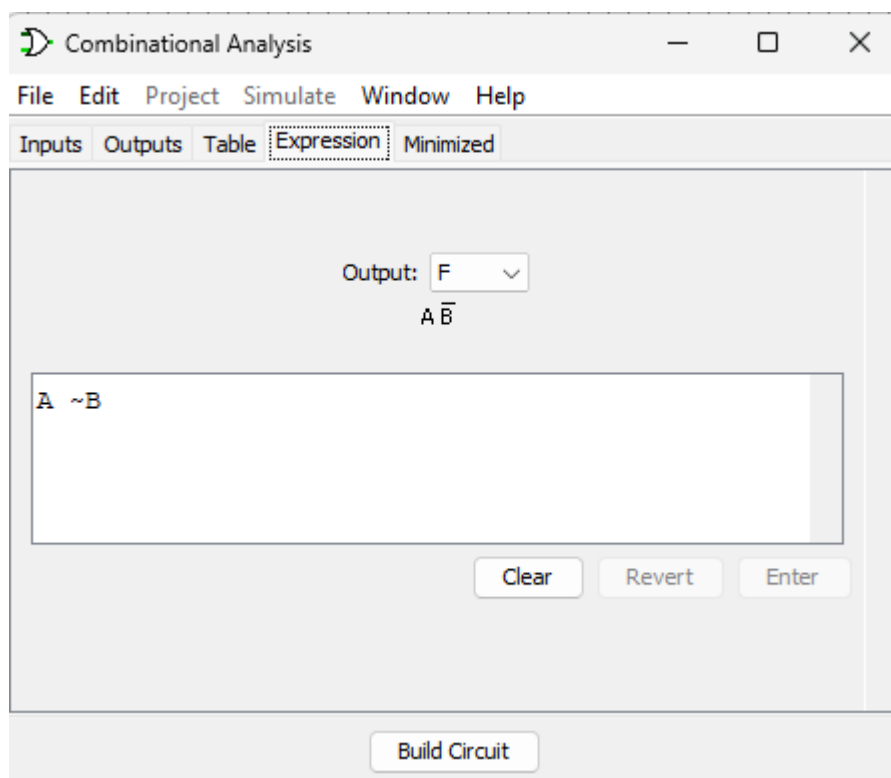
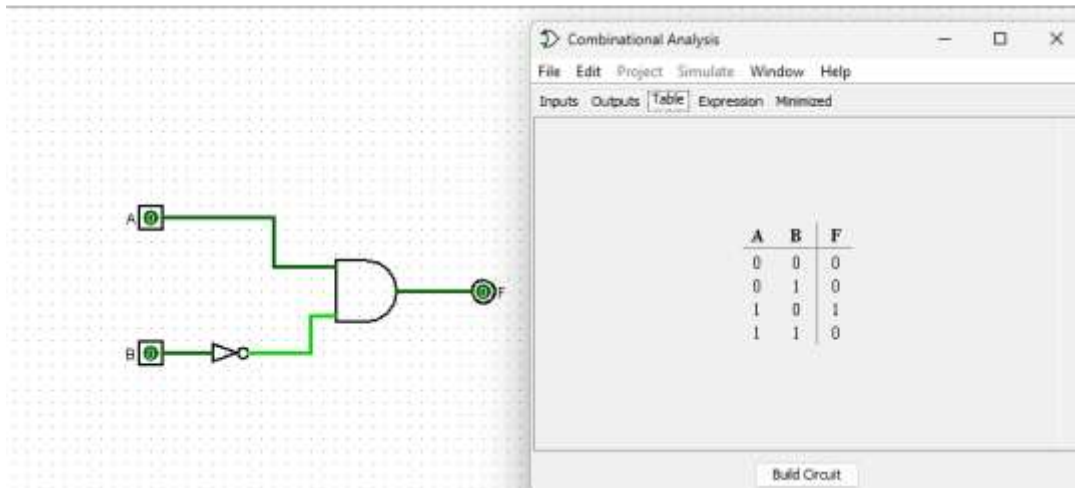


Figure 6. Diagram full adder.

3. Solved Exercises

1. Implementa las siguientes funciones usando puertas lógicas (1. NOT, AND y OR; 2. NAND; 3. NOR).

a. $F = A \cdot \bar{B}$



b. $G = (A + \bar{B}) \cdot (\overline{A + B})$

c. $H = (\bar{A} + A \cdot \bar{B}) \cdot (\overline{A + B}) \cdot C$

d. $H = (\overline{AC} + A\bar{B}\bar{C}) \cdot (\overline{AC + AB})$

2. Obtén e implementa la expresión de la función cuya tabla de verdad se da a continuación.

A	B	C	G
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

3. Obtén e implementa la expresión de la función cuya tabla de verdad se da a continuación .

A	B	C	D	H
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

4. Practical Problems

4. Se han instalado dos rótulos luminosos en la puerta de una consulta médica, uno con la leyenda “PASE”, y otro con la leyenda “ESPERE”. El primero debe encenderse solo si está el médico y no hay un paciente en el interior de la consulta. El segundo, cuando haya pasado un paciente. Se pide:
 - a. tabla de verdad de la función “P”, que nos indica el estado del cartel de “PASE”;
 - b. ídem para el rótulo “ESPERE”;
 - c. expresión algebraica de la función “P”;
 - d. ídem para la función “E”.
5. La alarma de una vivienda posee tres sensores: A, B y C. Dicha alarma debe activarse cuando por lo menos dos de los tres sensores estén activados. Se pide:
 - a. tabla de verdad de la función E (“estado de la alarma”);
 - b. simplifica la función algebraicamente;
 - c. simplifica la función usando un diagrama de Karnaugh;
 - d. implementa la función empleando cualquier tipo de puertas.

Learning Outcomes and Assessment Criteria

Contents	
1. Exploitation of microcomputer systems: • Computer architecture. • Introduction to microcomputer systems. • Representation of information.	
Learning outcomes	Evaluation criteria
LR 1. Evaluates computer systems identifying their components and characteristics.	a) The physical components of a computer system and its interconnection mechanisms have been recognized.

References:

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